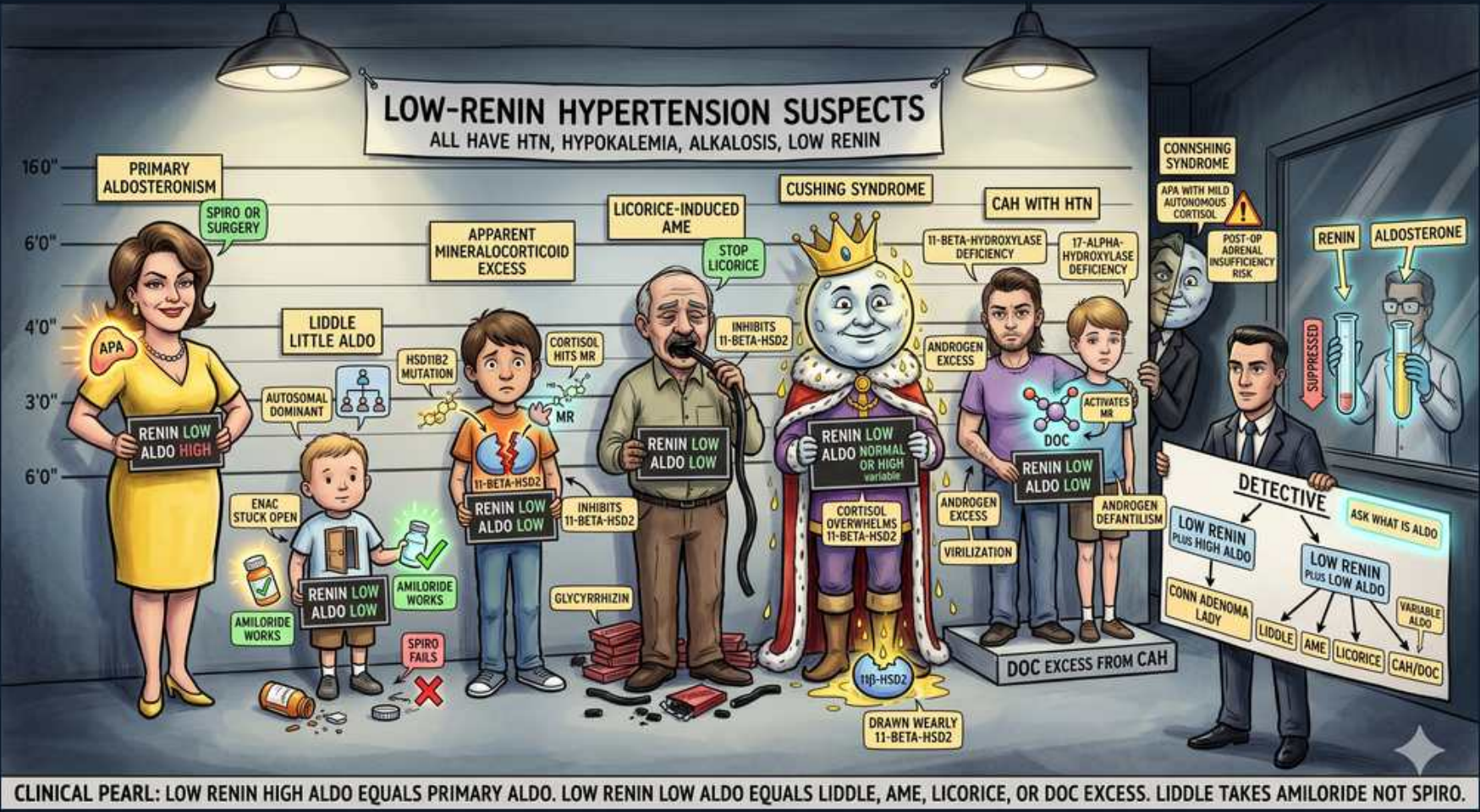


Primary Aldosteronism — The Low-Renin Hypertension Lineup



1. Shared signature

WHAT YOU SEE

Header: LOW-RENIN HYPERTENSION SUSPECTS — all have HTN, hypokalemia, alkalosis, low renin

WHAT IT ENCODES

Every cause in this lineup produces the same downstream picture: mineralocorticoid-driven HYPERTENSION, hypokalemia, metabolic alkalosis, and SUPPRESSED (low) renin. The mechanism is excess Na reabsorption through ENaC in the collecting duct, regardless of the upstream trigger. The single question that splits the lineup is the ALDOSTERONE level.

2. Primary aldosteronism (high aldo)

WHAT YOU SEE

Left woman APA, RENIN LOW ALDO HIGH, spiro or surgery

WHAT IT ENCODES

Primary aldosteronism (Conn adenoma / APA, or bilateral adrenal hyperplasia) is the ONLY suspect with LOW renin AND HIGH aldosterone → an elevated ARR. It is the most common cause of secondary HTN. Treat unilateral disease with adrenalectomy (potentially curative) and bilateral/non-surgical disease with an MR antagonist (spironolactone or eplerenone). High aldosterone is the defining discriminator versus everyone else in the line.

BOARD TRAP

Calling every low-renin HTN 'primary aldosteronism' — only this suspect has HIGH aldosterone. The others are LOW-renin LOW-aldo states.

3. Liddle syndrome

WHAT YOU SEE

Child LIDDLE LITTLE ALDO, ENaC stuck open, autosomal dominant, RENIN LOW ALDO LOW

WHAT IT ENCODES

Liddle syndrome is an autosomal DOMINANT gain-of-function mutation in the ENaC subunit (SCNN1B/SCNN1G) that prevents channel degradation, so ENaC is constitutively OPEN independent of aldosterone. The result is Na retention, HTN, hypokalemia, and metabolic alkalosis with LOW renin AND LOW aldosterone — it mimics aldosteronism with NO aldosterone. Suspect in young patients with a family history of severe HTN.

4. Liddle: amiloride not spiro

WHAT YOU SEE

AMILORIDE WORKS, spironolactone X over Liddle child

WHAT IT ENCODES

Because the defect is the ENaC channel itself (not the MR), Liddle responds to ENaC blockers — AMILORIDE or triamterene — and does NOT respond to spironolactone. Spironolactone blocks the MR, but in Liddle the MR is irrelevant since the channel is open on its own.

BOARD TRAP

Giving spironolactone for Liddle — it fails. The channel is the problem, not the receptor; use amiloride/triamterene.

5. Apparent mineralocorticoid excess (AME)

WHAT YOU SEE

Figure APPARENT MINERALOCORTICOID EXCESS, 11-beta-HSD2 mutation, cortisol hits MR, RENIN LOW ALDO LOW

WHAT IT ENCODES

Congenital AME is loss of 11β-HSD2, the enzyme that normally converts cortisol → inactive cortisone in the kidney to protect the non-selective MR. Without it, normal circulating CORTISOL floods and activates the MR, producing a mineralocorticoid effect with LOW renin and LOW aldosterone. A classic clue is a high serum/urinary cortisol-to-cortisone ratio. Treat with MR blockade (spironolactone) and/or dexamethasone to suppress endogenous cortisol.

6. Licorice-induced AME

WHAT YOU SEE

LICORICE-INDUCED AME, STOP LICORICE, inhibits 11-beta-HSD2

WHAT IT ENCODES

Glycyrrhetic/glycyrrhizic acid in natural (black) LICORICE — and in some chewing tobacco — pharmacologically INHIBITS 11β-HSD2, producing an ACQUIRED form of AME: cortisol again activates the MR → HTN, hypokalemia, alkalosis, low renin, low aldosterone. The treatment is simply to STOP the licorice. Always ask about licorice in unexplained low-renin/low-aldo HTN.

7. Cushing syndrome

WHAT YOU SEE

Crowned king CUSHING SYNDROME, cortisol overwhelms 11-beta-HSD2, drawn weary

WHAT IT ENCODES

In severe cortisol excess (especially ectopic ACTH / Cushing), the sheer quantity of cortisol OVERWHELMS the capacity of 11β-HSD2, so cortisol spills onto the MR and causes a mineralocorticoid effect → HTN and hypokalemia with low renin and low aldosterone. Marked hypokalemic alkalosis is more typical of ectopic ACTH than of a pituitary adenoma.

8. CAH with HTN

WHAT YOU SEE

Figures CAH WITH HTN: 11-beta-hydroxylase and 17-alpha-hydroxylase deficiency

WHAT IT ENCODES

The two congenital adrenal hyperplasia variants that cause HYPERTENSION are 11β-hydroxylase (CYP11B1) deficiency and 17α-hydroxylase (CYP17A1) deficiency. Both block cortisol synthesis but allow accumulation of the mineralocorticoid 11-deoxycorticosterone (DOC) → HTN, hypokalemia, low renin, low aldosterone. They differ on androgens (see next).

BOARD TRAP

The HTN-causing CAH is 11β- or 17α-hydroxylase deficiency. The COMMON CAH — 21-hydroxylase deficiency — causes salt WASTING with HYPOtension and HIGH renin, the opposite picture.

9. DOC excess from CAH

WHAT YOU SEE

Note DOC EXCESS FROM CAH, androgen excess/virilization vs androgen deficiency

WHAT IT ENCODES

11-deoxycorticosterone (DOC), a weak mineralocorticoid, builds up proximal to both enzyme blocks and drives the HTN/hypokalemia. Androgen direction splits them: 11 β -hydroxylase deficiency → DOC + androgen EXCESS (virilization, ambiguous genitalia in girls); 17 α -hydroxylase deficiency → DOC excess but androgen/estrogen DEFICIENCY (under-virilized boys, primary amenorrhea / absent puberty in girls).

10. Cushing → ectopic/iatrogenic

WHAT YOU SEE

Right CUSHING SYNDROME panel, ACTH-driven, post-op adrenal insufficiency risk

WHAT IT ENCODES

ACTH-dependent Cushing (especially ectopic ACTH from small-cell lung or carcinoid) and exogenous/iatrogenic glucocorticoid excess also produce low-renin HTN. After curative surgery for endogenous Cushing, anticipate adrenal insufficiency from the suppressed contralateral axis and replace glucocorticoids.

11. Detective: aldo decides

WHAT YOU SEE

DETECTIVE board: ASK WHAT IS ALDO — low renin high aldo vs low renin low aldo branches

WHAT IT ENCODES

The diagnostic fork: once renin is LOW, measure aldosterone. LOW renin + HIGH aldosterone = primary aldosteronism (confirm with ARR then salt-loading/saline-infusion non-suppression). LOW renin + LOW aldosterone = Liddle, AME (genetic), licorice, Cushing, or CAH (DOC excess) — distinguish these by history (licorice, steroids), cortisol/cortisone ratio, genetics, and the androgen phenotype.

12. Renin/aldo discriminator vials

WHAT YOU SEE

Two lab jars on the far right labeled RENIN and ALDOSTERONE

WHAT IT ENCODES

The two vials are the only labs you truly need to triage the lineup: renin and aldosterone. Renin is suppressed in ALL of them, so renin alone cannot discriminate — aldosterone is the decisive measurement. Remember to correct hypokalemia and stop interfering drugs (MRAs, ACEi/ARB) before drawing them for reliable results.

13. Clinical pearl

WHAT YOU SEE

Bottom banner

WHAT IT ENCODES

Bottom line: LOW renin + HIGH aldo = primary aldosteronism (MRA or surgery); LOW renin + LOW aldo = Liddle (amiloride), AME (11 β -HSD2 loss — congenital or licorice), Cushing, or DOC-excess CAH (11 β -/17 α -hydroxylase). Liddle takes amiloride, NOT spironolactone; 21-hydroxylase CAH does the OPPOSITE (salt wasting, high renin).